9.5 Making Change

WORKSHEET

The question

STEP I 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself How many ways are there to make n pence using only 2p and 1p coins? Why does the answer depend only on how many 2p coins you use?

STAGE 1 Ten pence

Ask yourself Organise the count by the largest coins. How many 10p coins can you use? Given that, how many 5p coins?

STAGE 2 Building up layer by layer

Ask yourself Let $b(n)$ be the number of ways to make n pence from 5p, 2p and 1p coins. How can you express $b(n)$ using the function a?

STAGE 3 **Twenty pence**

Pause and try Notice the structure: each new coin adds a layer, and each layer is a sum of the previous layer at spacings of the new coin's value. Can you see how a computer would carry out this calculation for £1, or £100?

What's really going on

Coins as geometric series

Here is the idea that turns this into algebra. The number of 5p coins used can be $0, 1, 2, \ldots$, contributing $0, 5, 10, \ldots$ pence. Record this as the series

$$1 + x^5 + x^{10} + x^{15} + \cdots = \frac{1}{1 - x^5}.$$

Do the same for each coin and multiply:

$$\frac{1}{(1-x)(1-x^2)(1-x^5)(1-x^{10})(1-x^{20})} = \sum_n w(n)x^n.$$

When the brackets are multiplied out, each term x^n arises from choosing one term from each bracket, that is, a number of each coin, with total value n. So $w(n)$, the coefficient of x^n, is exactly the number of ways to make n pence. The table in Stage 2 was simply multiplying out this product one bracket at a time. This is the generating function idea of Section 9.4, and multiplication is where it shows its power: products of generating functions count combinations of choices.

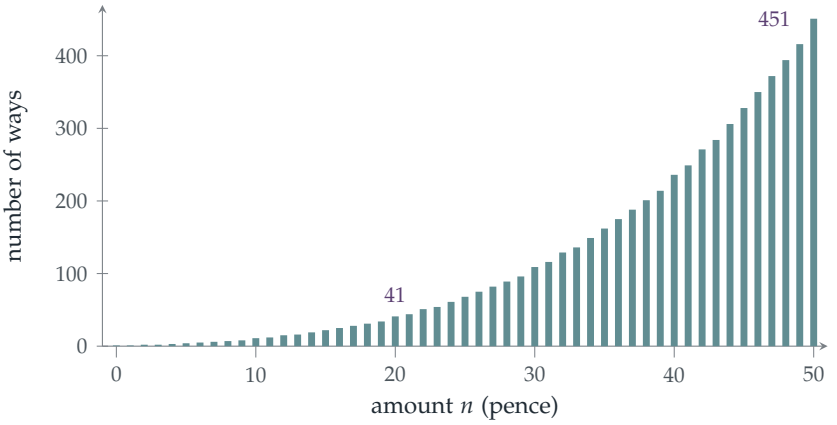


Figure 9.5.1 The number of ways to make n pence from 1p, 2p, 5p, 10p, 20p and 50p coins. The jumps at multiples of 5 and 10 show where a new coin first becomes usable. With six coins the count grows roughly like n^5: in general, like a polynomial of degree one less than the number of coins.

Partitions

What if every whole number is allowed as a coin? Then the number of ways to write n as a sum of positive integers, ignoring order, is the *partition number* $p(n)$. For example $p(4) = 5$, since $4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$. Euler saw in 1748 that

$$\sum_{n\geq 0} p(n)x^n = \frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)\cdots},$$

and used it to discover remarkable identities. One of them: the number of ways to write n as a sum of *distinct* parts equals the number of ways to write it as a sum of *odd* parts. (The proof is a single line of algebra with generating functions: try it in the exercises.)

Hardy, Ramanujan and $p(200)$

The partition numbers grow fast: $p(10) = 42$, $p(100) = 190\,569\,292$. In 1918 G. H. Hardy and Srinivasa Ramanujan found an astonishing formula for $p(n)$, an infinite series whose first term is

$$p(n) \approx \frac{1}{4n\sqrt{3}}\, e^{\pi\sqrt{2n/3}}.$$

To test it, Percy MacMahon computed $p(200) = 3\,972\,999\,029\,388$ by hand, using Euler's identities. The first few terms of Hardy and Ramanujan's series gave the right answer to within a fraction of one. Ramanujan also discovered that $p(5n+4)$ is always divisible by 5, $p(7n+5)$ by 7, and $p(11n+6)$ by 11. Explaining why there are no more such patterns took until the twenty-first century.

Try this next

1 **Two kinds of six.** Try STEP II 2012, Question 1 (step.maths.org/questions/12-s2-q1). It counts with products of geometric series. At the end, describe a coin-counting problem that the coefficients answer.

2 **Distinct and odd.** Show that $(1+x)(1+x^2)(1+x^3)\cdots$ is the generating function for partitions into distinct parts. Use $1+x^k = \frac{1-x^{2k}}{1-x^k}$ to show that it equals $\frac{1}{(1-x)(1-x^3)(1-x^5)\cdots}$, and deduce Euler's theorem. Check it for $n=6$.

3 **A formula.** Show that the number of ways to make n pence using only 1p, 2p and 5p coins is the nearest integer to $\frac{(n+4)^2}{20}$. (Check it first for small n.)

4 **Order matters.** In how many ways can you make n pence by putting 1p and 2p coins into a slot one at a time, if the order matters? (You have met this sequence before.)

Further reading

- **George Andrews & Kimmo Eriksson, *Integer Partitions*** (Cambridge University Press). A short, readable introduction for undergraduates.
- **George Pólya, "On picture-writing"**, *American Mathematical Monthly* 63 (1956). Pólya's classic explanation of generating functions using the problem of changing a dollar (there are 292 ways).
- **Robert Kanigel, *The Man Who Knew Infinity*** (Abacus). The life of Ramanujan, and his partnership with Hardy.

The full guided solution is in Section 9.5:
https://beyondstep.keithmcnulty.org/ch09/making-change.html